Proposed Solution

Date	09 July 2026
Team ID	AIML-HDI-2026
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This proposal aims to make Human Development Index estimation fast and interactive by replacing the manual UNDP formula workflow with a trained hybrid neural network exposed through a live web dashboard, improving both accuracy and accessibility for policy-oriented decision-making. Key features include a dual-branch regression model and a real-time prediction dashboard.

Project Overview

Field	Details
Objective	To predict a country's Human Development Index score and development tier in real time using a hybrid deep-learning model trained on key socio-economic indicators.
Scope	Comprehensively models HDI using primary UNDP-aligned indicators (life expectancy, schooling, GNI) plus auxiliary numerical indicators to build a robust and efficient prediction system.

Problem Statement

Field	Details
Description	Current reliance on manual, formula-heavy HDI computation prevents fast, exploratory, or predictive use of socio-economic indicator data.
Impact	Solving this enables faster, data-driven policy discussion and easier communication of development standing to non-specialist audiences.

Proposed Solution

Field	Details
Approach	A dual-branch hybrid neural network trained with the AdamW optimizer and MSE loss to regress the HDI score, with tier classification derived by threshold rule.
Key Features	• Hybrid regression model separating core and auxiliary indicators • GPU-accelerated data cleaning (cuDF) • Real-time web dashboard for prediction (Flask + HTML/CSS) • Publicly accessible via LocalTunnel during the notebook session

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	GPU used for training and preprocessing	Dual NVIDIA Tesla T4 GPUs (Kaggle Notebook environment)
Memory	RAM specification	Kaggle default runtime allocation (≥ 13 GB)
Storage	Disk space for datasets, models, and logs	Kaggle working directory (session-scoped)
Software		
Frameworks	Python frameworks used	PyTorch, Flask
Libraries	Additional libraries	cuDF, pandas, NumPy, scikit-learn, matplotlib, seaborn
Development Environment	IDE / Notebook platform	Kaggle Notebook (Jupyter-based)
Data		
Data	Data source	Kaggle — Human Development Index (Full) dataset